

Sarbasish Chowdhury

+91 9903729897 | sarbasishchowdhury.252is033@nitk.edu.in | linkedin.com/in/sarbasish | github.com/sarbasish | leetcode.com/sarbasish | portfolio/sarbasish

EDUCATION

National Institute of Technology Karnataka, Surathkal <i>Master of Technology, Computer Science and Engineering (Information Security)</i> CGPA: 8.26	Karnataka, India Aug. 2025 – Present
B P Poddar Institute of Management and Technology <i>Bachelor of Technology, Information Technology</i> CGPA: 9.04	Kolkata, India Aug. 2021 – Jul. 2025

EXPERIENCE

Programmer Analyst Trainee <i>Cognizant Technology Solutions</i> - Trained and worked on Automation Testing using Selenium. - Applied Core Java, DSA, and SQL to develop test scripts and validate backend logic. - Experienced in Functional Testing, Agile methodology, Jira, Maven, Git, and API Automation.	May. 2025 – Jul. 2025 Kolkata, India
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PROJECTS

TrustNetAI <i>Python, PyTorch, Transformers, BERT, SHAP, SHA-256</i> - Designed and developed TrustNetAI, a hybrid intelligent informatics framework integrating BERT-based fake news detection, SHAP explainability, and blockchain-based trust validation. - Built an end-to-end misinformation analysis pipeline including data preprocessing, transformer-based classification, token-level explainability, and SHA-256 hash-based verification mechanisms. - Fine-tuned BERT models on FakeNewsNet dataset and evaluated system performance using accuracy, precision, recall, and F1-score metrics.	Feb. 2026 – Present
Secure Query DB <i>Python, FastAPI, PostgreSQL, AES-256, HMAC, RBAC, Blockchain</i> - Designed and developed a secure distributed database framework enabling privacy-preserving query execution over encrypted and sharded databases using AES-256-GCM encryption and searchable encryption techniques. - Implemented trapdoor-based secure search with HMAC-generated keyword tokens, multi-keyword AND queries, deterministic shard routing, and role-based access control (RBAC) for secure query coordination. - Built an end-to-end architecture integrating FastAPI query coordination, PostgreSQL shards, blockchain-based audit logging, and Streamlit-based real-time query visualization and monitoring.	Jan. 2026 – Apr. 2026
DeepViNE <i>Python, PyTorch, NetworkX, DQN, CNN</i> - Implemented a complete DeepViNE system from scratch, including physical network modeling, virtual network generation, resource mapping, reward functions, and RL environment. - Built a Dueling DQN agent with CNN-based state encoding and action-space design for VNE problem. - Reproduced evaluation metrics (blocking probability, revenue, utilization).	Oct. 2025 – Dec. 2025

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, HTML/CSS
Databases: MySQL, MongoDB
Frameworks: React, JDBC, REST API, Selenium
Developer Tools: Git, GitHub, IntelliJ, Eclipse, VS Code, Postman, Jira
Libraries: NumPy, Matplotlib, PyTorch
Core CS Concepts: DSA, OOP, OS, DBMS, CN
Areas of Interest: Cryptography, Information Security, Gen AI

ACHIEVEMENTS

- Qualified **GATE 2025 (CS)**: 99.34 percentile, AIR 1171, Score 708